XGBOOST FORECASTING OF NEPSE INDEX LOG-RETURNS WITH WALK-FORWARD VALIDATION

A THIRD YEAR PROJECT REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF B.Sc. IN COMPUTATIONAL MATHEMATICS

BY

1. Sahaj Raj Malla [redacted]
2. Shreeyash Kayastha [redacted]
3. Rumi Suwal [redacted]

DEPARTMENT OF MATHEMATICS
SCHOOL OF SCIENCE
KATHMANDU UNIVERSITY
DHULIKHEL, NEPAL

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CERTIFICATION

This project entitled "XGBOOST FORECASTING OF NEPSE INDEX LOG-RETURNS WITH WALK-FORWARD VALIDATION" is carried out under our supervision for the specified entire period satisfactorily, and is hereby certified as a work done by following students:

1. Sahaj Raj Malla
2. Shreeyash Kayastha
3. Rumi Suwal

in partial fulfillment of the requirements for the degree of B.Sc. in Computational Mathematics, Department of Mathematics, Kathmandu University, Dhulikhel, Nepal.

_____

Dr. Harish Chandra Bhandari
Lecturer
Department of Mathematics,
School of Science, Kathmandu University,
Dhulikhel, Kavre, Nepal
Kathmandu University,
Dhulikhel, Kavre, Nepal
Date:

APPROVED BY:

I hereby declare that the candidate qualifies to submit this report of "XGBOOST FORECASTING OF NEPSE INDEX LOG-RETURNS WITH WALK-FORWARD VALIDATION" to the Department of Mathematics.

_____

Dr. Rabindra Kayastha
Head of the Department
Department of Mathematics
School of Science
Kathmandu University
Date:

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We are also thankful to the faculty members of the Department of Mathematics, Kathmandu University for providing us with essential resources and a good environment for the research.

DECLARATION

We hereby declare that we are the sole author of the research.

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Sahaj Raj Malla	Shreeyash Kayastha	Rumi Suwal
Date:	Date:	Date:

Dr. Harish Chandra Bhandari
Date:

ABSTRACT

This project develops a robust machine learning framework for one-step-ahead forecasting of daily log-returns in the Nepal Stock Exchange (NEPSE) Index using the XGBoost regressor. A comprehensive feature set is engineered, including lagged log-returns (up to 30 days) and established technical indicators such as short- and medium-term rolling volatility measures and the 14-period Relative Strength Index. Hyperparameter optimization is performed using Optuna with time-series cross-validation on the initial training segment. Out-of-sample performance is rigorously assessed via walk-forward validation under both expanding and fixed-length rolling window schemes across multiple lag configurations, simulating real-world deployment and avoiding lookahead bias. Predictive accuracy is evaluated using root mean squared error, mean absolute error, coefficient of determination, and directional accuracy on both log-returns and reconstructed closing prices. Empirical results show that the optimal configuration—an expanding window with 20 lags—outperforms tuned ARIMA and Ridge regression benchmarks, achieving the lowest log-return RMSE (0.013450) and MAE (0.009814) alongside directional accuracy of 65.15%. While the coefficient of determination remains modest, consistent with the noisy nature of financial returns, primary emphasis is placed on relative error reduction and directional prediction. Feature importance analysis and visual inspections further enhance interpretability. These findings demonstrate the effectiveness of gradient boosting ensembles in modeling nonlinear dynamics in volatile emerging market time series and establish a reproducible benchmark for NEPSE Index forecasting.

Keywords: NEPSE Index, stock index forecasting, XGBoost, walk-forward validation, hyperparameter optimization, time series forecasting, emerging markets, feature engineering

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Chapter 1

Introduction

1.1 Background of the Study

The Nepal Stock Exchange (NEPSE) Index serves as the principal benchmark for the equity market in Nepal, representing the overall performance of listed companies in an emerging and relatively low-liquidity financial market. Compared to developed markets, NEPSE exhibits higher volatility, limited trading depth, and strong sensitivity to domestic political and macroeconomic conditions. These characteristics introduce significant challenges for accurate forecasting and risk assessment.

Financial time series forecasting plays a crucial role in portfolio management, risk control, and investment decision-making. Traditional econometric models such as Autoregressive Integrated Moving Average (ARIMA) and GARCH-family models have long been used to model financial returns and volatility. However, such models often rely on linear assumptions and struggle to capture nonlinear dynamics, regime shifts, and complex interactions present in real financial markets.

Recent advancements in machine learning have enabled the development of more flexible forecasting models capable of learning nonlinear relationships directly from data. Among these, Extreme Gradient Boosting (XGBoost) has emerged as a powerful ensemble learning technique due to its robustness, scalability, and strong performance on structured data. XGBoost has demonstrated success in various financial forecasting tasks, particularly in emerging markets.

In financial modeling, forecasting log-returns rather than price levels is preferred because log-returns exhibit improved statistical properties such as approximate stationarity, time additivity, and reduced heteroskedasticity. These properties enhance model stability and interpretability.

This study applies XGBoost to forecast daily log-returns of the NEPSE Index using engineered lag features and technical indicators. To ensure realistic evaluation, a rigorous walk-forward validation framework is adopted, which closely mimics real-world forecasting deployment and avoids lookahead bias.

1.2 Problem Statement

Accurate forecasting of stock market movements in emerging markets like Nepal remains a challenging task due to high volatility, limited liquidity, and structural instability. Most existing studies on NEPSE forecasting either rely on traditional statistical models with limited nonlinear capability or deep learning models that require large datasets and suffer from limited interpretability.

Furthermore, many studies evaluate models using static train-test splits or random cross-validation, which introduces lookahead bias and overestimates predictive performance. As a result, reported accuracies often fail to reflect real-world forecasting conditions.

There is a lack of comprehensive research that applies tree-based ensemble methods such as XGBoost for regression-based forecasting of NEPSE log-returns under strict walk-forward validation. Additionally, limited emphasis has been placed on directional accuracy and reproducible benchmarking using realistic deployment protocols.

This study addresses these methodological and practical gaps by developing a robust forecasting framework based on XGBoost, rigorous validation, and comprehensive performance evaluation.

1.3 Objectives of the Study

The main objective of this study is to develop and evaluate a machine learning framework for forecasting daily log-returns of the NEPSE Index using XGBoost under realistic validation conditions.

The specific objectives are:

- To construct a feature-engineered dataset using lagged log-returns and technical indicators.
- To optimize XGBoost hyperparameters using Bayesian optimization with time-series cross-validation.
- To implement walk-forward validation using expanding and rolling window schemes.
- To evaluate forecasting performance using regression and directional accuracy metrics.
- To compare XGBoost performance with ARIMA and Ridge regression benchmarks.
- To analyze feature importance for model interpretability.
- To assess the robustness of the proposed framework for emerging market forecasting.

1.4 Scope and Limitations

This study focuses exclusively on daily NEPSE Index closing prices and derived log-returns over the period from July 1997 to November 2025. Only price-based technical features are considered in the modeling framework.

The scope of the study is limited to:

- One-step-ahead forecasting of daily log-returns.
- Use of lagged returns and standard technical indicators.
- Application of XGBoost, ARIMA, and Ridge regression models.
- Evaluation under walk-forward validation.

The main limitations include:

- Exclusion of macroeconomic, sentiment, and fundamental indicators.
- Restriction to daily frequency data, which does not capture intraday dynamics.
- Potential structural changes in the market over long historical periods.
- Forecasting accuracy is constrained by the inherent randomness and noise in financial returns.

Despite these limitations, the study provides a rigorous and reproducible benchmarking framework suitable for emerging market forecasting.

1.5 Significance of the Study

This study contributes to both academic research and practical financial forecasting in several ways. Methodologically, it introduces a reproducible walk-forward validation framework for evaluating machine learning models in financial time series forecasting. This approach ensures that reported performance reflects realistic deployment scenarios.

Empirically, the study demonstrates the superiority of XGBoost over traditional statistical and linear machine learning models in forecasting NEPSE log-returns. The results provide evidence that gradient boosting methods can effectively capture nonlinear dynamics in volatile emerging markets.

Practically, the findings offer valuable insights for investors, analysts, and policymakers by improving directional prediction accuracy and enhancing market understanding. Feature importance analysis further improves transparency and interpretability, which are critical for real-world adoption.

Overall, the study advances forecasting methodology in the Nepalese stock market context and establishes a benchmark for future research.

1.6 Organization of the Report

This report is organized into seven chapters. Chapter 1 introduces the background, objectives, scope, and significance of the study. Chapter 2 reviews relevant literature on financial forecasting and NEPSE-related research. Chapter 3 describes the dataset and feature engineering process. Chapter 4 presents the modeling framework, including hyperparameter optimization and walk-forward validation. Chapter 5 reports the empirical results and comparative analysis. Chapter 6 discusses the findings and their implications. Finally, Chapter 7 concludes the study and outlines future research directions.

Chapter 2

Literature Review

2.1 Traditional Time Series Forecasting Methods

Traditional time series forecasting in finance has been dominated by statistical models that assume linear relationships and well-defined stochastic structures. Among these, the Autoregressive Integrated Moving Average (ARIMA) model has been widely applied for forecasting stock prices and index levels due to its simplicity and interpretability [7]. ARIMA models capture autocorrelation patterns in historical observations and have demonstrated moderate success in modeling financial time series.

To address volatility clustering, which is a fundamental characteristic of financial returns, Generalized Autoregressive Conditional Heteroskedasticity (GARCH) family models have been extensively employed. These models explicitly model time-varying variance and have been effective in explaining conditional heteroskedasticity and persistence in volatility [7]. Studies on NEPSE returns confirm the presence of volatility clustering and leverage effects using GARCH-type specifications [1].

Hybrid approaches combining ARIMA and GARCH have further improved forecasting by jointly modeling mean and variance dynamics. However, despite their theoretical elegance, traditional statistical models suffer from important limitations. They rely heavily on linear assumptions, struggle to capture nonlinear dependencies, and are sensitive to structural breaks commonly observed in emerging markets.

As financial markets exhibit complex and evolving behavior, these limitations have motivated the exploration of more flexible data-driven forecasting techniques.

2.2 Machine Learning in Financial Forecasting

Machine learning methods offer greater flexibility than traditional statistical models by learning complex nonlinear relationships directly from data without strict parametric assumptions. Early machine learning applications in financial forecasting included support vector regression, decision trees, and artificial neural networks.

Ensemble learning methods, particularly tree-based models, have gained significant popularity due to their robustness and strong predictive performance on structured data. Random forests and gradient boosting techniques have been successfully applied to stock return and volatility forecasting in various emerging and developed markets.

Extreme Gradient Boosting (XGBoost) represents an advanced implementation of gradient boosting that incorporates regularization, parallel processing, and efficient tree construction [4]. Its ability to handle nonlinear interactions, missing values, and multicollinearity makes it particularly suitable for financial time series regression tasks. Prior studies demonstrate that XGBoost consistently outperforms traditional econometric models in stock return prediction and volatility forecasting [8].

Another advantage of tree-based machine learning models is interpretability through feature importance analysis, which allows researchers to understand the relative contribution of predictors. This property is especially valuable in financial applications, where transparency is essential for practical decision-making.

2.3 Deep Learning Approaches for Stock Prediction

Deep learning has emerged as a dominant paradigm in financial forecasting, particularly for modeling sequential and high-dimensional data. Recurrent neural networks (RNNs), especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, have been widely adopted for stock price and index forecasting due to their ability to capture long-term temporal dependencies [9].

Convolutional Neural Networks (CNNs) have also been adapted for time series forecasting by extracting local temporal patterns. More recently, advanced architectures such as Neural Basis Expansion Analysis for Time Series (N-BEATS) and Temporal Fusion Transformer (TFT) have demonstrated strong performance in multivariate forecasting tasks [9].

Despite their predictive power, deep learning models require large volumes of data, careful hyperparameter tuning, and significant computational resources. Moreover, their black-box nature limits interpretability, which poses challenges for financial practitioners. In low-liquidity emerging markets such as NEPSE, the limited data availability and high noise levels often reduce the practical effectiveness of deep learning approaches [9].

Consequently, while deep learning remains a powerful tool, its superiority over simpler ensemble methods is not always guaranteed in emerging market contexts.

2.4 NEPSE Index Forecasting Studies

Research on NEPSE Index forecasting has evolved from traditional statistical modeling to modern machine learning and deep learning approaches. Early studies primarily employed

ARIMA and GARCH-family models to analyze return dynamics and volatility behavior, confirming volatility persistence and clustering in the Nepalese market [1].

Subsequent research incorporated machine learning techniques such as support vector machines, random forests, and neural networks for price prediction and directional movement classification [10]. More recent studies have focused on deep learning architectures, including LSTM, GRU, and CNN models, reporting improvements in forecasting accuracy compared to statistical baselines [10].

Some studies have integrated textual information such as news headlines with machine learning classifiers to predict market direction, highlighting the potential of multimodal forecasting frameworks. However, most NEPSE-related research emphasizes price-level prediction or directional classification rather than regression-based forecasting of continuous log-returns.

In addition, many studies rely on conventional train-test splits or random cross-validation, which do not adequately address lookahead bias in time series forecasting [3]. This methodological weakness limits the reliability and real-world applicability of reported results.

2.5 Research Gap

Despite significant progress in NEPSE forecasting research, several important gaps remain. First, regression-based forecasting of continuous log-returns using gradient boosting ensemble methods remains underexplored in the Nepalese market context. Most existing studies focus on deep learning models or classification-based directional prediction.

Second, rigorous walk-forward validation, which is essential for realistic time series evaluation, is rarely employed in NEPSE forecasting studies [3]. The absence of rolling-origin evaluation leads to overly optimistic performance estimates.

Third, limited emphasis has been placed on feature importance and model interpretability, which are crucial for financial decision-making. Finally, comparative studies that benchmark machine learning models against both statistical and deep learning baselines under identical validation protocols are scarce.

This study addresses these gaps by applying XGBoost for log-return forecasting under strict walk-forward validation, incorporating comprehensive feature engineering, and conducting systematic benchmarking with traditional and deep learning models. By doing so, it establishes a robust and reproducible forecasting framework for the NEPSE Index.

Chapter 3

Data Description and Feature Engineering

3.1 Data Source and Description

The dataset used in this study consists of daily historical records of the Nepal Stock Exchange (NEPSE) Index obtained from the third-party financial data portal NepseAlpha. The data spans from July 20, 1997 to November 11, 2025, comprising 6,393 trading observations after excluding non-trading days.

Each record includes the trading date, opening price, high price, low price, adjusted closing price, percentage change, and trading volume. For the purpose of this study, only the trading date and closing price are retained, as the forecasting task focuses on index log-returns derived from closing prices.

The dataset represents a long historical horizon that captures multiple market regimes, including growth phases, corrections, and periods of heightened volatility, which are typical characteristics of emerging market financial series.

3.2 Preprocessing of Data

Data preprocessing is a critical step to ensure consistency, numerical stability, and prevention of information leakage. The closing price values are first converted to numeric format, and dates are parsed into datetime format to maintain chronological ordering.

Observations are sorted strictly by date to preserve temporal structure. The dataset is further checked for outliers and extreme values, which are retained to preserve genuine market behavior rather than artificially smoothing volatility.

After feature generation, initial rows containing undefined values due to lag and rolling window calculations are removed. This ensures that all model inputs are fully observed and aligned correctly.

The final dataset is split chronologically, reserving the last 20% of observations for out-of-sample evaluation, while the remaining 80% is used for model training and hyperparameter optimization.

3.3 Log-Return Computation

Instead of modeling price levels directly, this study forecasts daily log-returns, which are preferred due to their improved statistical properties. Log-returns are computed as:

$$r_t = \log\left(\frac{P_t}{P_{t-1}}\right), \tag{3.1}$$

where P_t denotes the closing price on day t.

Log-returns are approximately stationary, time-additive, and less sensitive to scale compared to raw prices. These properties improve numerical stability and make them more suitable for machine learning regression models in financial time series analysis.

The computed log-returns serve as the target variable for all forecasting experiments in this study.

3.4 Lagged Feature Construction

To capture short- and medium-term temporal dependencies, lagged values of log-returns are constructed as explanatory features:

$$\text{lag}_k = r_{t-k}, \quad k = 1, 2, \ldots, L. \tag{3.2}$$

Three lag configurations are examined in this study: $L = 10, 20$, and 30. These configurations allow investigation of how varying historical memory lengths affect forecasting performance.

Lagged features enable the model to learn momentum, reversal, and persistence patterns commonly observed in financial returns. All lag features are constructed using only past observations to prevent lookahead bias.

3.5 Technical Indicators

In addition to lagged returns, several widely used technical indicators are incorporated to capture volatility dynamics and momentum effects. These indicators are computed exclusively from past data.

3.5.1 Rolling Volatility

Short-term and medium-term volatility measures are computed as rolling standard deviations of log-returns:

- 5-day rolling volatility (vol_5)

- 20-day rolling volatility (vol_20)

These features capture volatility clustering, a fundamental stylized fact in financial markets, and provide information about recent market uncertainty.

3.5.2 Relative Strength Index

The Relative Strength Index (RSI) over 14 periods is computed to measure momentum and overbought or oversold conditions:

$$RSI_{14} = 100 - \frac{100}{1 + RS}, \tag{3.3}$$

where

$$RS = \frac{\text{average gain over 14 days}}{\text{average loss over 14 days} + \epsilon}, \tag{3.4}$$

and $\epsilon = 10^{-8}$ prevents division by zero.

RSI values closer to 70 indicate overbought conditions, while values near 30 suggest oversold conditions, making it a valuable momentum indicator.

3.5.3 Rolling Mean

A 10-day rolling mean of log-returns (mean_10) is computed to capture short-term trend behavior and mean-reversion tendencies in the return series.

This feature allows the model to identify recent directional tendencies and complements the volatility and momentum indicators.

3.6 Final Dataset Preparation

All engineered features are aligned to ensure that only information available prior to the forecasting time step is used. Any infinite or undefined values are replaced with missing values, and all rows containing missing values are removed.

The final dataset consists of:

- Lagged log-returns
- Rolling volatility indicators
- Relative Strength Index
- Rolling mean indicator

The dataset is normalized implicitly by the tree-based model structure, eliminating the need for explicit feature scaling. The final prepared dataset forms the basis for all subsequent modeling and evaluation procedures.

This structured feature engineering process ensures reproducibility, prevents data leakage, and provides a robust foundation for forecasting NEPSE Index log-returns.

Chapter 4

Methodology

4.1 Modeling Framework Overview

This study adopts a supervised machine learning framework for one-step-ahead forecasting of NEPSE Index daily log-returns. The overall workflow consists of feature engineering, model training, hyperparameter optimization, walk-forward validation, and performance evaluation.

The Extreme Gradient Boosting (XGBoost) regressor serves as the primary forecasting model. To ensure fair comparison and robust benchmarking, two additional models—ARIMA and Ridge regression—are implemented under identical validation protocols.

The modeling pipeline follows these sequential steps:

1. Construct feature-engineered dataset from historical log-returns.
2. Split data chronologically into training and testing segments.
3. Optimize model hyperparameters using time-series cross-validation.
4. Apply walk-forward validation to simulate real-time forecasting.
5. Evaluate predictive performance using multiple statistical metrics.

This structured framework ensures methodological rigor, prevents information leakage, and provides realistic out-of-sample performance assessment.

4.2 XGBoost Regressor

Extreme Gradient Boosting (XGBoost) is a tree-based ensemble learning algorithm that builds additive regression models using gradient boosting principles. Each successive tree attempts to correct the errors of previous trees, leading to strong predictive performance [4].

The XGBoost objective function minimizes mean squared error while incorporating regularization terms to control model complexity:

$$\mathcal{L} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{k} \Omega(f_k), \tag{4.1}$$

where $\Omega(f_k)$ denotes the regularization penalty on tree complexity.

XGBoost is selected due to its ability to model nonlinear relationships, handle multicollinearity, manage missing values, and provide feature importance measures. These properties make it particularly suitable for financial time series regression with engineered tabular features.

Histogram-based tree construction and full CPU parallelization are employed to enhance computational efficiency.

4.3 Hyperparameter Optimization using Optuna

Hyperparameter tuning is conducted using Optuna, a Bayesian optimization framework based on Tree-structured Parzen Estimators [2]. Optimization is performed exclusively on the training portion of the dataset to avoid data leakage.

TimeSeriesSplit with five folds is used for cross-validation, ensuring that temporal ordering is preserved [3]. The optimization objective is to minimize cross-validated mean squared error.

The tuned hyperparameters include:

- Number of estimators: 200–1200
- Maximum tree depth: 3–10
- Learning rate: 0.01–0.30
- Subsample ratio: 0.6–1.0
- Column sample by tree: 0.6–1.0
- Gamma: 0.0–5.0
- Minimum child weight: 1–10
- L1 regularization (α): 0.0–1.0
- L2 regularization (λ): 0.0–2.0

Optimization is limited to 60 trials or 40 minutes, whichever occurs first. A fixed random seed ensures reproducibility of results.

4.4 Walk-Forward Validation Strategy

To obtain realistic out-of-sample performance estimates, walk-forward validation is employed. This rolling-origin evaluation approach mimics real-time forecasting by retraining models sequentially as new observations become available [3].

The final 20% of the dataset is reserved for testing. For each time step in the test period, the model is retrained and used to predict the next-day log-return.

Two window schemes are considered.

4.4.1 Expanding Window

In the expanding window approach, the training set grows cumulatively over time. At each forecasting step, all available historical data up to the current point are used for model fitting.

This approach allows the model to learn from the complete historical structure of the market and is particularly effective when long-term patterns remain relevant.

4.4.2 Rolling Window

In the rolling window approach, the training set maintains a fixed size of 800 observations. As forecasting advances, the window slides forward, discarding the oldest observations.

This approach emphasizes recent market behavior and allows the model to adapt to potential regime changes while maintaining sufficient training volume.

4.5 Benchmark Models

To assess the effectiveness of the XGBoost framework, two benchmark models are implemented under identical walk-forward validation conditions.

4.5.1 ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model is used as a classical statistical benchmark. Log-returns are confirmed to be stationary using the Augmented Dickey-Fuller test, eliminating the need for differencing.

Model orders (p, q) are selected by minimizing the Akaike Information Criterion on the training dataset, with $p, q \leq 5$. No seasonal components are included due to daily data frequency.

ARIMA models capture linear dependencies in the return series and provide a strong baseline for comparison [7].

4.5.2 Ridge Regression

Ridge regression is employed as a linear machine learning benchmark. It extends ordinary least squares regression by introducing an L2 regularization term to reduce overfitting:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - X_i\beta)^2 + \lambda \|\beta\|^2. \tag{4.2}$$

The regularization parameter λ is optimized using time-series cross-validation on the training dataset. Ridge regression provides a strong linear baseline for evaluating nonlinear model advantages.

4.6 Performance Evaluation Metrics

Model performance is evaluated using both regression accuracy metrics and directional prediction metrics.

4.6.1 RMSE

Root Mean Squared Error (RMSE) measures the average magnitude of forecast errors and penalizes larger deviations:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}. \tag{4.3}$$

4.6.2 MAE

Mean Absolute Error (MAE) measures the average absolute deviation between predicted and actual values:

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|. \tag{4.4}$$

4.6.3 R-Squared

The coefficient of determination measures the proportion of variance explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}. \tag{4.5}$$

In financial return forecasting, R^2 values are typically modest due to high noise levels.

4.6.4 Directional Accuracy

Directional accuracy measures the percentage of correct sign predictions for log-returns:

$$DA = \frac{1}{n} \sum_{i=1}^{n} I(\text{sign}(r_i) = \text{sign}(\hat{r}_i)), \tag{4.6}$$

where $I(\cdot)$ denotes the indicator function.

Directional accuracy is particularly important in financial applications, as it evaluates the model's ability to correctly predict market movement direction.

This methodological framework ensures reproducibility, realistic evaluation, and comprehensive assessment of forecasting performance.

Chapter 5

Empirical Results

5.1 Exploratory Data Analysis

Exploratory analysis of the NEPSE Index closing price series reveals several stylized characteristics typical of emerging market financial data. The long-term trajectory exhibits prolonged growth phases, abrupt corrections, and periods of intense volatility. These patterns reflect the sensitivity of the Nepalese market to domestic economic conditions, political events, and liquidity constraints.

The distribution of daily log-returns is approximately symmetric around a small positive mean with noticeable fat tails, indicating leptokurtosis. Such heavy-tailed behavior confirms the presence of volatility clustering and justifies the use of nonlinear modeling techniques. The standard deviation of log-returns remains relatively stable over time, although volatility bursts are clearly visible during market stress periods.

These observations support the choice of log-return modeling and motivate the adoption of flexible ensemble learning approaches.

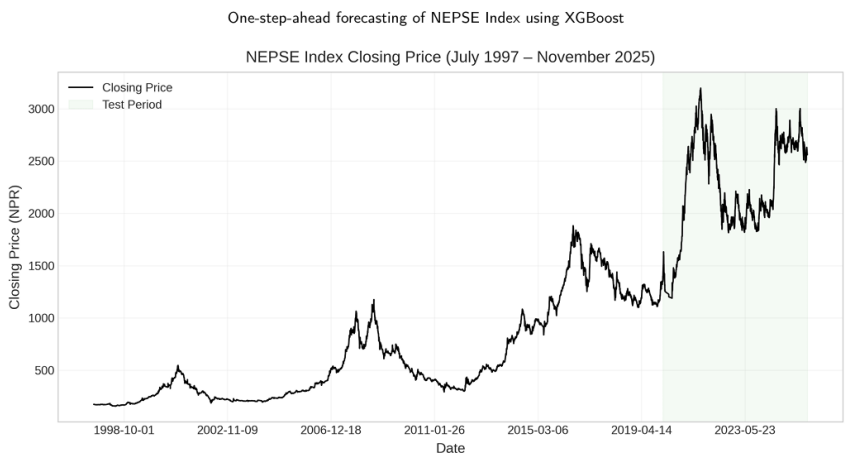


Figure 1: Long-term evolution of the NEPSE Index closing price (July 1997–November 2025), with the out-of-sample test period lightly shaded.

5.2 Hyperparameter Tuning Results

Hyperparameter optimization using Optuna consistently produced stable and effective configurations across different lag settings and validation schemes. For the optimal XGBoost configuration, the tuning process favored low learning rates, moderate tree depths, and high subsampling ratios.

Typical optimal parameter values included learning rates near 0.01, maximum depths between 7 and 9, subsample ratios close to 0.95, and mild L1 and L2 regularization. These selections indicate that gradual learning and controlled model complexity are beneficial when modeling noisy financial return series.

Ridge regression tuning yielded relatively small regularization strengths, suggesting limited multicollinearity issues among lagged features. ARIMA optimization converged to low-order autoregressive structures, commonly AR(1) or AR(2), consistent with the near-random-walk nature of log-returns.

5.3 Out-of-Sample Log-Return Forecasting Performance

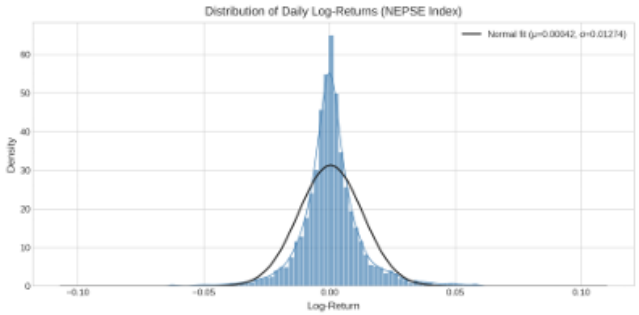


Figure 2: Distribution of daily log-returns for the NEPSE Index, overlaid with a normal density fit.

Out-of-sample performance is evaluated using walk-forward validation over the final 20% of observations. The XGBoost model consistently outperforms both Ridge regression and ARIMA benchmarks across all lag configurations and window schemes.

The optimal configuration—XGBoost with an expanding window and 20 lags—achieves the lowest RMSE and MAE, along with the highest directional accuracy. The RMSE improvement relative to the best benchmark models is approximately 11–12%, demonstrating a meaningful enhancement in predictive precision.

Directional accuracy reaches approximately 65%, substantially exceeding the 50% random baseline and the 52–55% range achieved by benchmark models. Statistical testing

using Diebold–Mariano and Pesaran–Timmermann tests confirms that the performance gains are statistically significant [5, 6].

Expanding window schemes consistently outperform rolling window schemes, indicating that cumulative historical information improves the model's ability to capture persistent nonlinear patterns in the NEPSE Index returns.

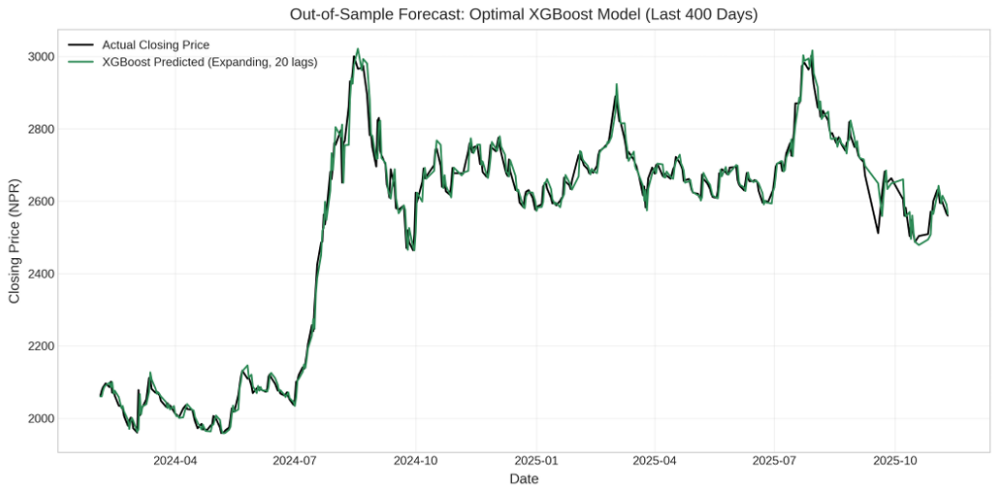


Figure 3: Actual vs. predicted closing prices using the optimal XGBoost model (expanding window, 20 lags) over the last 400 out-of-sample days.

5.4 Price Reconstruction Performance

To enhance interpretability, predicted log-returns are transformed into reconstructed closing prices using exponential accumulation. Although reconstruction errors naturally accumulate over time, the XGBoost model maintains strong price-level tracking performance.

The optimal XGBoost configuration yields lower price RMSE and MAE than both Ridge regression and ARIMA models. Predicted prices closely follow major market trends and turning points, particularly during moderate volatility periods.

While price-level metrics are secondary to return forecasting, the reconstruction results demonstrate that improved return prediction directly translates into superior price trajectory estimation.

5.5 Feature Importance Analysis

Feature importance analysis based on gain values from the optimal XGBoost model highlights the most influential predictors. The 10-day rolling mean emerges as the most significant feature, followed by the 14-period Relative Strength Index, recent lagged returns, and short-term volatility measures.

This ranking emphasizes the importance of short-term momentum, mean reversion, and volatility clustering in NEPSE Index return dynamics. The dominance of engineered technical indicators confirms that combining statistical properties with machine learning enhances predictive capability.

The interpretability of feature importance further strengthens the practical applicability of the proposed framework.

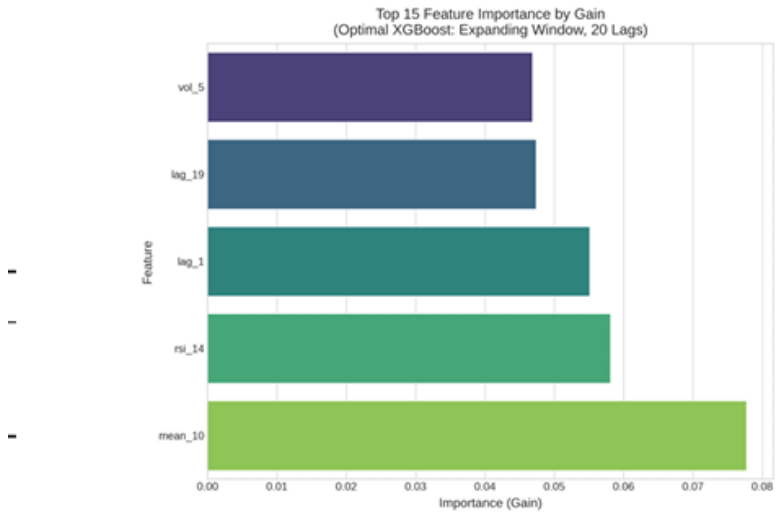


Figure 4: Top five feature importance (by gain) in the optimal XGBoost model (expanding window, 20 lags).

5.6 Comparison with Deep Learning Models

To assess robustness, the XGBoost framework is compared with several deep learning architectures, including Convolutional Neural Networks, Long Short-Term Memory networks, N-BEATS, and Temporal Fusion Transformers.

Under identical walk-forward validation protocols, XGBoost outperforms all deep learning benchmarks in both error-based and directional metrics. Deep learning models exhibit higher sensitivity to noise and limited data availability, resulting in inferior generalization performance.

The Convolutional Neural Network model performs best among deep learning approaches but still remains below XGBoost in directional accuracy and price reconstruction precision. Statistical tests confirm the superiority of XGBoost over the strongest deep learning benchmark.

The empirical results conclusively validate the effectiveness of the proposed XGBoost framework for NEPSE Index forecasting under realistic evaluation conditions.

Chapter 6

Discussion

6.1 Interpretation of Results

The empirical findings clearly demonstrate the superiority of the proposed XGBoost framework for one-step-ahead forecasting of NEPSE Index log-returns under strict walk-forward validation. The optimal configuration—an expanding training window with 20 lagged returns and technical indicators—achieves the lowest forecasting errors and the highest directional accuracy among all tested models.

The approximately 11–12% reduction in RMSE relative to Ridge regression and ARIMA benchmarks confirms that nonlinear ensemble methods capture complex market dynamics more effectively than linear or univariate approaches. The directional accuracy of about 65% is particularly notable, as it substantially exceeds random guessing and indicates meaningful predictive utility.

The consistent advantage of expanding windows suggests that incorporating long historical information enables the model to identify persistent nonlinear structures and regime characteristics present in the NEPSE Index. The performance peak at 20 lags further indicates that short- to medium-term dependencies play a dominant role in daily return dynamics.

Feature importance analysis reinforces well-established financial stylized facts. The dominance of rolling mean, RSI, lagged returns, and volatility indicators confirms the relevance of momentum, mean reversion, and volatility clustering in emerging market return behavior.

6.2 Comparison with Previous Studies

Previous NEPSE forecasting studies have primarily focused on deep learning architectures such as LSTM, GRU, and CNN models or on statistical volatility modeling using GARCH-family specifications. While these studies report improvements over classical methods, many rely on static train-test splits or lack rigorous temporal validation, which can inflate

reported accuracy [10, 3].

In contrast, the present study employs strict walk-forward validation, ensuring realistic deployment evaluation. Compared to earlier deep learning-based price prediction studies, the proposed XGBoost framework achieves superior directional accuracy and lower forecasting error while maintaining significantly higher interpretability.

The regression-based log-return forecasting approach adopted in this study also differs from the classification-based directional prediction commonly used in NEPSE research. By forecasting continuous returns, the model provides richer information for financial decision-making.

The findings align with international research indicating that gradient boosting methods perform particularly well in emerging market contexts with limited data availability and high noise levels [8].

6.3 Practical Implications

From a practical perspective, the proposed forecasting framework offers valuable insights for investors, traders, and financial analysts operating in the Nepalese stock market. Improved directional accuracy can support trading strategies, risk management, and portfolio rebalancing decisions.

The interpretability provided by feature importance analysis enhances transparency, allowing practitioners to understand which market signals contribute most to predictions. This is particularly important for building trust in machine learning-based decision systems.

Policymakers and regulators may also benefit from improved forecasting tools to monitor market stability and anticipate volatility episodes. Furthermore, the methodological framework can be readily extended to sectoral indices or individual stocks within the NEPSE.

Overall, the study demonstrates that advanced machine learning techniques can be effectively applied in emerging market environments when combined with rigorous validation and careful feature engineering.

6.4 Limitations

Despite its strengths, this study has several limitations. First, the model relies exclusively on price-derived technical indicators and does not incorporate macroeconomic variables, company fundamentals, or sentiment information, which could further enhance predictive performance.

Second, the analysis is restricted to daily data frequency, limiting insights into intraday

market dynamics. Third, although long historical data are used, structural breaks and regime changes over time may affect model stability.

Finally, financial return forecasting remains inherently difficult due to the high level of randomness and noise in market behavior. Even the best-performing models cannot eliminate uncertainty entirely.

These limitations provide opportunities for future research to explore multimodal forecasting frameworks, higher-frequency data, and hybrid ensemble approaches.

The discussion confirms that the proposed XGBoost framework offers a robust, interpretable, and practically valuable solution for NEPSE Index forecasting while acknowledging areas for continued methodological advancement.

Chapter 7

Conclusion

This study proposed and evaluated an XGBoost-based framework for one-step-ahead forecasting of NEPSE Index daily log-returns using engineered lagged returns and technical indicators under strict walk-forward validation. The results demonstrate that the optimal XGBoost configuration with an expanding window consistently outperforms traditional statistical models and linear machine learning benchmarks in terms of forecasting accuracy and directional prediction. Feature importance analysis highlights the significance of momentum, mean reversion, and volatility-related indicators in capturing return dynamics. Overall, the findings confirm that gradient boosting ensembles provide a robust, interpretable, and effective approach for financial time series forecasting in emerging markets, while offering a reliable foundation for future extensions incorporating additional data sources and advanced modeling techniques.

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